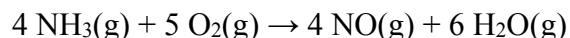


NOTE: Please keep a copy of your HW#12 to study from. It may not be graded before the final exam.

[1] In class we worked an example of burning ammonia and calculating the exit temperature of the gases.



The feed stream of 4 mol/s ammonia (NH_3) and 6 mol/s of oxygen (O_2) enters the reactor at 200°C and the product contain no ammonia and leaves at an unknown temperature $T_{\text{out}}(^{\circ}\text{C})$. Because of the limited range of applicability of the heat capacity relations we found two solutions for the exit temperature; namely, 2657 K and 3626 K. Rework the energy balance using better expression for the heat capacities. Go to:

<http://webbook.nist.gov/chemistry/>

and select Name under Search Options. Fill in the name of the compound and its phase and retrieve the data. You should use the Shomate equations for the ideal gas, constant-pressure heat capacity. Plot the Shomate equation for water from 25°C to 6000°C and compare with the expression given in Murphy on the graph. Show that the Shomate equation can be used down to 298 K and give reasonable results. What is the exit temperature when calculated using the Shomate equation? Explain why you now find only one solution.

[2] In class we began an example problem for the dehydrogenation reaction,



where the equilibrium constant is defined as $K_p(\text{atm}) = \frac{X_{\text{C}_2\text{H}_4} X_{\text{H}_2}}{X_{\text{C}_2\text{H}_6}} P$ The equilibrium constant has been found

experimentally to be $K_p(T) = \kappa \exp\{-\varepsilon/T\}$ where $\kappa = 7.28 \times 10^6 \text{ atm}$ and $\varepsilon = 17,000 \text{ K}$. The heat of reaction at 1273 K is +145.1 kJ/mol for the reaction as written. The heat capacities in J/(mol·K) are given approximately as:

$$(C_p)_{\text{C}_2\text{H}_4} = 9.419 + 0.1147T$$

$$(C_p)_{\text{H}_2} = 26.90 + 0.004167T \quad \text{where } T \text{ is in Kelvin}$$

$$(C_p)_{\text{C}_2\text{H}_6} = 11.35 + 0.1392T$$

Pure ethane is fed into a continuous constant-pressure adiabatic reactor at 1273 K and pressure $P(\text{atm})$, the products emerge at $T(\text{K})$ and $P(\text{atm})$, and the residence time of the reactor is long enough that the products are at equilibrium.

In class we proved that

$$f = \left(\frac{K_p}{P + K_p} \right)^{1/2} \quad \text{and} \quad f = \frac{1}{1 + \phi(T)} \quad \text{where}$$

$$\phi(T) = \frac{\Delta \hat{H}_r(1273\text{K}) - \int_T^{1273\text{K}} [(C_p)_{\text{C}_2\text{H}_4} + (C_p)_{\text{H}_2}] dT}{\int_T^{1273\text{K}} [(C_p)_{\text{C}_2\text{H}_6}] dT}$$

Combine these results to give an expression in the form $\psi(T) = 0$. Solve the equation for different values of $P(\text{atm}) = 0.01, 0.05, 0.1, 0.5, 1.0, 5.0$ and determine the fractional conversion f . Plot T versus P and f versus P in two separate plots.